Faculty of Science, Engineering and Built Environment

SEJ102 Electrical Systems Engineering Project

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

This is a project and design-based unit which introduces students to the fundamentals of electrical systems, covering three main topics; electricity basics including capacitance, resistance, inductance and EMF; electric circuits; and semi-conductor devices. The unit will also introduce students to elements of professional practice including OHS, professional ethics and sustainability in the context of engineering projects. Students will work in groups, using their knowledge gained through classes, studios and practical experiences, the concepts covered in the seminar sessions, as well as knowledge gained from self-directed learning to complete their design of an electrical systems project meeting specified requirements. Students will communicate the considerations taken in their design in a variety of forms including written assessments tasks and a group presentation.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Scott Adams
Campus: Geelong Waurn Ponds
Email: scott.adams@deakin.edu.au
Phone: +61 3 522 73772
Other members of the team and how to contact them

Name: Dean Corva
Campus: Geelong Waurn Ponds
Email: d.corva@deakin.edu.au
Phone: +61 3 522 72453
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- The wide variety of learning resources.

- The practical studio sessions.
- The electronics project assignments.
- Having clearly defined expectations.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- We have updated the assessment rubrics, and we have improved the studio tasks.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Demonstrate discipline specific knowledge analysis and skills relating to electrical fundamentals.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Apply a general engineering methodology that is common to most disciplines of professional engineering.	GLO4: Critical thinking
ULO3	Apply a holistic design process that incorporates principles of sustainability to solutions to be applied within built and natural environments.	GLO4: Critical thinking GLO5: Problem solving
ULO4	Effectively communicate in a professional manner using common methods applied in business communication.	GLO2: Communication
ULO5	Demonstrate an awareness of OHS, professional ethics and sustainability in the context of Engineering projects.	GLO8: Global citizenship

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must receive at least 40% in the group presentation, and at least 50% in the individual portfolio.

Brief summary of the hurdle requirement	Rationale
To be eligible to obtain a pass in this unit, students must receive at least 40% in the group oral presentation, and at least 50% in the individual portfolio.	The Group Presentation requires students to demonstrate oral communication of technical information, and this ability is linked to ULO4 and GLO2. The Individual Portfolio is a significant assessment task with various components and is linked to all ULOs.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted

after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Online tests**
Brief description of assessment task	Students are required to apply their understanding of electrical concepts in the completion of two online tests, which will evaluate student's understanding of the key concepts covered in the unit.
Detail of student output	Two short answer 3-hour online tests undertaken individually by students via the unit site.
Grading and weighting (% total mark for unit)	20% (2 x 10% per test) Marked
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1- Demonstrate discipline specific knowledge, analysis and skills relating to electrical fundamentals.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1- through student ability to recall and apply their understanding of electrical fundamentals to a series of questions.
How and when you will receive feedback on your work	Students will receive feedback on the tests after they have been marked by an assessor. The feedback will include any questions answered incorrectly and provision of full worked solutions. This will be provided within one to two weeks of completing the test.
When and how to submit your work	Students will complete these tests via the unit site. 1. **Test 1:** Friday 22 August 2025 (Week 6), 00.01am – 8pm (AEST) 2. **Test 2:** Friday 19 September 2025 (Week 10), 00.01am – 8pm (AEST) The online tests will be open for the entire period to give all students the opportunity to undertake the test.

Summative assessment task 2

Title	**Group research and design proposal**
Brief description of assessment task	This assessment task requires students to work in allocated groups to articulate the proposed design of their project and submit it as a written Design Proposal.
	Working as a group students need to draw upon their knowledge gained through the various learning activities for the unit, and propose a design which satisfies the requirements as set out by the teaching team.
Detail of student output	Student groups are required to submit a single written Design Proposal according to the assessment specification. The Design Proposal will include the proposed project design, discussion on progress to date in completing the required project and a safety plan relating to undertaking the project. It will also include an element of assessment related to professional ethics and sustainability as applicable to the project design. The design proposal should be between 5 and 10 pages in length. It is the group's responsibility to ensure that the workload is managed equitably.
Grading and weighting (% total mark for unit)	10% Marked

This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Demonstrate discipline specific knowledge, analysis and skills relating to electrical fundamentals. ULO2 - Apply a general engineering methodology that is common to most disciplines of professional engineering. ULO3 - Apply a holistic design process that incorporates principles of sustainability to solutions to be applied within built and natural environments. ULO4 - Effectively communicate in a professional manner using common methods applied in business communication. ULO5 - Demonstrate an awareness of OHS, professional ethics and sustainability in the context of Engineering projects.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – through student ability to apply their understanding of electrical fundamentals to their group's proposed design. GLO2 - through student ability to effective communicate their proposed design in a written proposal. GLO4 - through demonstration of critical judgement in the proposed design. GLO5 - through proposal of a design satisfying the project requirements. GLO8 - through inclusion of a safety plan and demonstration of an understanding of OHS, professional ethics and sustainability.
How and when you will receive feedback on your work	Students will have the opportunity to seek regular feedback during seminar sessions. The design proposal will be marked against an assessment rubric and feedback will be given once assessed generally within 1-2 weeks.
When and how to submit your work	Friday 29 August 2025, 8pm (AEST) Week 7. The Design Proposal will be submitted in electronic format via the unit site. All submissions must be in single pdf format (You may need to learn how to combine multiple pdfs to do this). All reports are to be submitted with the following naming Convention. UnitCode_YEAR_Trimester_LastName_FirstName_StudentID

Summative assessment task 3

Title	**Group Presentation**
Brief description of assessment task	This assessment task requires students groups to give a succinct oral presentation outlining the main considerations taken in the proposed design of their project. Student groups need to incorporate appropriate visual aids and other information to augment their oral presentation.
Detail of student output	Student groups are required to give a 10 minute online oral presentation on their proposed design. Group members need to share the role of presenter evenly and this will be specifically assessed as part of the assessment rubric.
Grading and weighting (% total mark for unit)	10% Marked
This task assesses your achievement of these Unit Learning Outcome(s)	ULO4 - Effectively communicate in a professional manner using common methods applied in business communication.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO2 - through student ability to orally and visually communicate the group's design considerations.
How and when you will receive feedback on your work	Feedback on the project presentation will be provided both online and via a marked rubric, during the scheduled presentation session. The feedback will include a component on the technical considerations of the design.

When and how to submit your work	All presentations will be conducted via Zoom in Week 10 at a time to be determined by the Unit Chair. Specific dates and times will be provided on the unit site. Note: Oral presentation slides and videos must be uploaded to the assessment dropbox prior to your presentation.

Summative assessment task 4

Title	**Individual portfolio**
Brief description of assessment task	This assessment task is completed individually and requires students to submit a portfolio containing evidence of their efforts during the trimester. Details of content will be covered in detail during week 1. The portfolio needs to include your contributions to weekly problems, laboratories, studios, seminars and lecture work. Details of content and assessment criteria can be found in the rubric.
Detail of student output	This is an individual assessment task. Students are required to create an individual portfolio of their trimester's work and submit as a single PDF via the unit site.
Grading and weighting (% total mark for unit)	60% Graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Demonstrate discipline specific knowledge, analysis and skills relating to electrical fundamentals. ULO2 - Apply a general engineering methodology that is common to most disciplines of professional engineering. ULO3 - Apply a holistic design process that incorporates principles of sustainability to solutions to be applied within built and natural environments. ULO4 - Effectively communicate in a professional manner using common methods applied in business communication. ULO5 - Demonstrate an awareness of OHS, professional ethics and sustainability in the context of Engineering projects.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – through student ability to present and justify their understanding of electrical fundamentals to their group's proposed design. GLO2 - through student ability to discuss and reflect on their proposed project in a written report. GLO4 – through student's demonstrated ability to discuss design considerations taken as well as reflection on design choices made. GLO5 - through demonstration of a well formed solution as evidenced in the portfolio. GLO8 – through presented discussion of professional ethics and sustainability as its relates to the completed project design.
How and when you will receive feedback on your work	Students have the opportunity to seek regular feedback during the seminar sessions. The portfolio will be marked against an assessment rubric and students will receive feedback via the unit site within one to two weeks of submission.
When and how to submit your work	Week 12, Monday 29 September 2025, 8pm (AEST). All submissions must be in single pdf format (You may need to learn how to combine multiple pdfs to do this). All reports are to be submitted with the following naming Convention. UnitCode_YEAR_Trimester_LastName_FirstName_StudentID

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 3 hour practical experience (studio) session per week, 1 x 1 hour seminar per week.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 1 x 1 hour seminar per week. Students are required to attend and participate in project oriented design-based learning activities at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site.

Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

Learning materials shall be provided in the form of online video content of approximately 2 hours per week (provided in the unit site), plus 3 hours of studio exercises and 1 hour of seminars per week. Additional seminar Zoom sessions may be scheduled depending on demand.

Studios and projects shall be in the form of electronic submission. All students shall be expected to complete electronic studio and project work in teams using electronic media to collaborate with team members and organise work distribution and resources, as necessary.

Presentations shall be completed via Zoom and timetabled typically in week 10. Electronic submission shall be made via the unit site on or before the due date/time.

There will be a series of on-campus activities for both on campus and online students. The online students will undertake these activities as part of intensive week. On-campus students will complete these tasks each week as part of the on-campus studio sessions.

All students should register for the on-campus studio sessions.

Note that the evening streaming session is weekly and will contain the seminars for each week.

The on-campus studios will consist of a series of introductory exercises including:

- Laboratory equipment introduction (Power supply operation, multimeters, Digital Scopes)
- Soldering (Hand soldering and surface mount soldering
- Hand tool safety
- Design prototyping

Other activities may be scheduled and will be advised.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
Prescribed text(s): Floyd & Buchla, 2025, Electronics Fundamentals: Circuits Devices & Applications, 9th global edition, (or 8PNIE), Pearson Education.

The texts and reading list for SEJ102 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

This unit has a prescribed textbook (see above).

Note: This textbook is also available as an eText from Pearson Australia.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

In addition to the required textbook, the Unit Team will also prepare learning resources accessible through the unit site. This includes video recordings for the classes delivered in this Unit (accessible to students of all campuses, Geelong, Burwood) and other resources as appropriate.

Student Equipment Purchases

Students are advised to acquire basic hand tools for soldering and electronics/electrical work as well as testing.

This non-compulsory equipment will be discussed in the first week and would cost approximately $100.

- A good multi-meter with fine tipped probes
- Precision side cutters (100mm)
- Precision long nose pliers (100mm)
- Precision Screwdrivers

Optional: soldering station with fine tip and variable temperature control, 2 Channel Oscilloscope, 12V 3A bench top power supply.
This equipment will be used at various levels throughout the unit.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025

Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	7 July 2025	Electrical Systems Fundamentals and Project related topics	Skills building exercises online and practical	
2	14 July 2025	Electrical Systems Fundamentals and Project related topics		
3	21 July 2025	Electrical Systems Fundamentals and Project related topics	Skills building exercises online and practical	
4	28 July 2025	Electrical Systems Fundamentals and Project related topics		
5^	4 August 2025	Electrical Systems Fundamentals and Project related topics		
6	18 August 2025	Electrical Systems Fundamentals and Project related topics		**Assessment 1: Online Test 1** due 22 August
7	25 August 2025	Electrical Systems Fundamentals and Project related topics		**Assessment 2: Group Research and Design Proposal** due 29 August
8	1 September 2025	Electrical Systems Fundamentals and Project related topics		
9	8 September 2025	Electrical Systems Fundamentals and Project related topics		
10	15 September 2025	Electrical Systems Fundamentals and Project related topics		**Assessment 1: Online Test 2** due 19 September **Assessment 3: Group oral presentation** (dates and times to be confirmed)
11#	22 September 2025	Electrical Systems Fundamentals and Project related topics		
12	29 September 2025			**Assessment 4: Individual Portfolio** due 29 September Study period
13-14	6 October 2025			End-of-unit assessment and exam period

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed